

Data Mining - Project

Come up with an analysis for a set of data of your choice. It is recommended that the dataset is consistent enough to allow for meaningful insights to be discovered in your investigation.

The data that you choose can be of any of the types of data we have worked with in the laboratory:

- text data: social media posts, reviews, books, articles, research papers, movie subtitles etc.
- tabular data: any dataset(s) containing text / numeric / date-time / binary data or any combination of those

The analysis should make use of the knowledge gained from the laboratory. However, you are free to expand this and use any data mining methods you find suitable as long as they are properly understood and explained in your project.

Be creative in the use of what you learned in the laboratory.

Goal

The goal of the project is to apply what you have studied at the lab to come up with a meaningful investigation which you can successfully present to an audience.

Teams

You will work in teams of 2.

Technical Topics

Choose and apply 6 of the topics below in your project:

- **Data Cleaning & Preprocessing (Mandatory)**
- TF-IDF
- Language Models (applied to your dataset)
- N-grams
- PoS tags
- NER
- Web scraping

- Topic modelling
- Sentiment Analysis
- EDA / Data Visualization
- Feature importance analysis / Feature selection / PCA
- Information Retrieval
- Recommendation Systems
- Apply & Evaluate Supervised Model 1 (include your fine tunings)
- Apply & Evaluate Supervised Model 2 (include your fine tunings)
- Apply & Evaluate Unsupervised Model 1 (include your fine tunings)
- Apply & Evaluate Unsupervised Model 2 (include your fine tunings)

Choose the topics you find best suit the data you have chosen for your project (e.g. don't perform topic analysis on numeric data).

Make sure you share meaningful insights for these topics and present them adequately, $\frac{2}{3}$ of the jury grade is based on the presentation of your findings.

Turning In

You will submit an archive file containing the following 2 documents:

- python notebook (with comments)
- presentation (max 7 minutes / team, **presentation will be interrupted after 8 minutes**)

You will then present your analysis during the scheduled exam date and time.

Grading

$\frac{1}{2}$ Graded by the jury (made up of professor + lab assistants)

$\frac{1}{2}$ Graded by other teams

Both of the grades will be computed as:

- $\frac{1}{3}$ Clarity and Effectiveness of the Presentation
- $\frac{1}{3}$ Technical Rigor and Creativity
- $\frac{1}{3}$ Insightfulness of the Analysis

Clarity and Effectiveness of the Presentation:

- The presentation must communicate the insights clearly, using visualizations and simple language when appropriate.
- Time management matters. Get to the point. Avoid reading code or overloading slides with text.

We're evaluating how well you make the audience understand and care about your findings

Technical Rigor and Creativity

- Projects should show careful data preparation, and appropriate selection of methods for the problem.
- We reward good judgment in method choice (e.g. not applying sentiment analysis blindly).
- Going beyond the lab work is expected — please make sure you understand what you're doing and can explain it clearly. Creative or original applications are appreciated.

It's not about complexity – it's about using the right tools, in the right way, to tell a compelling data story.

Insightfulness of the Analysis

- The project should go beyond describing the data. We expect clear, well-supported insights that reveal non-obvious patterns, trends, or implications.
- For each method used, there should be an interpretation: *What does this result tell us? Why does it matter?*

Be aware, a graph or model without interpretation will not score well.